

Algebraic Structures

①

$$(N, +), (Z, -), (R, +, -)$$

Algebraic Structures

$$N = \{1, 2, 3, \dots\} \quad \text{Operation: } +$$

*) Algebraic Structures follow some properties.

(I) Definition of Properties $\rightarrow$

(a) Binary Operation $\rightarrow$ let X be a non-empty set.

A binary operation $*$ on X is denoted as $x * y$, which is performed on x and y elements of X to form another element.

(b) Closure law $\rightarrow$ let X be a non-empty set.

The set X is called closed under $*$, if it satisfies the following properties $\rightarrow$

$$x \in X, y \in X \Rightarrow x * y \in X$$

eg. $(N, +) \quad 3 + 2 = 5 \checkmark$

$(X, +) \quad X = \{-2, 0, 1, 2\} \quad 2 + 2 = 4 \notin X$

(c) Associative Law $\rightarrow$ The binary operation $*$ is said to be associative on the set X if for all

$$x, y, z \in X,$$

$$x * (y * z) = (x * y) * z$$

eg. $(N, +)$

$$3 + (2 + 5) = (3 + 2) + 5$$
$$\underline{3 + 7} = \underline{5 + 5}$$
$$\underline{10} = \underline{10} \checkmark$$

(d) Existence of Identity Element $\rightarrow$

If there exists an element $e \in X$, such that for all $x \in X$,

$$x * e = e * x = x$$

$e \rightarrow$ Identity Element.

eg. $(\mathbb{Z}, +)$

$$e = 0$$

[Identity element is Unique for a Set]

(e) Existence of Inverse Element

If for each element $x \in X$, there exists an element $y \in X$ such that,

$$x * y = y * x = e$$

then y is the inverse of x

eg. $(\mathbb{R}, +)$

$$5 + (?) = 0$$

$$5 + (-5) = 0$$

[Every element should have inverse] -5 is the inverse for 5

(f)

Commutative Law

The binary operation $*$ is said to satisfy the commutative law, if $\forall x, y \in X$,

$$x * y = y * x$$

eg. $(\mathbb{R}, *)$

$$5 * 4 = 4 * 5$$

$$20 = 20$$

(II) Algebraic Structures $\rightarrow$

- $\rightarrow$ Let X be a set and $*$ be a binary operation defined on the set X .
- $\rightarrow$ The binary operation on the given set represents a structure between the elements of the set.
- $\rightarrow$ Thus the Algebraic Structure is represented as $\rightarrow$
 $(X, *)$

(a) Groupoid $(X, *)$

$\rightarrow$ ① Closure $\forall x, y \in X, x * y \in X$

eg. $(\mathbb{R}, -)$ $\begin{matrix} 2, 3 \\ (2-3) = -1 \in \mathbb{R} \checkmark \end{matrix}$

(b) Semi-Groups $(X, *)$

① Closure $\forall x, y \in X, x * y \in X$

② Associativity $\forall x, y, z \in X$
 $(x * y) * z = x * (y * z)$

eg. $(\mathbb{Z}, +)$

Closure $\checkmark$
Associativity $\checkmark$

$$\begin{matrix} (2+3)+5 & = & 2+(3+5) \\ 10 & & 10 \end{matrix}$$

$(\mathbb{Z}, -)$

Closure $\checkmark$
Associativity $\times$

$$\begin{matrix} (3-2)-8 & \neq & 3-(2-8) \\ -7 & & 3-(-6) = 9 \end{matrix}$$

(c) Monoid $(X, *)$

- ① Closure
- ② Associativity
- ③ Has identity element in X ,
 $\forall x \in X, x * e = e * x = x$

eg. $(\mathbb{Z}, +)$

Closure ✓

Associativity ✓

$e = 1$

eg $2 \times 1 = 1 \times 2 = 2$

Identity ✓

(d) Groups $(X, *)$

- ① Closure
- ② Associativity
- ③ Identity
- ④ Inverse Element,
 $\forall x \in X, \exists x^{-1} \in X$ such that
 $x * x^{-1} = e$

eg $(\mathbb{R}, \times)$

$$x \times \frac{1}{x} = 1$$

$$x^{-1} = \frac{1}{x}$$

$\nexists x=0, x^{-1}$ does not exist/undefined
 $\therefore 0$ does not have inverse in $\mathbb{R}$,
 $\therefore (\mathbb{R}, \times)$ is not a group

(e)

Abelian Group $(X, *)$

⑤

- ① Closure
- ② Associativity
- ③ Identity
- ④ Inverse
- ⑤ Commutative, i.e.

$$x * y = y * x \quad \forall x, y \in X$$

eg. $(\mathbb{R}, +)$

$$2 + 3 = 3 + 2 \quad \checkmark$$

Q

Let $(R, *)$ be an Algebraic Structure, where binary operation $*$ is defined in the following table. Identify the Algebraic Structure in each.

1.	$a + b$
2.	$a \cdot b$
3.	$\min(a, b)$
4.	$a - b$
5.	$a + b - ab$
6.	a^b

Solution $R(-\infty, \infty)$ open

		Closure	Associativity	Identity	Inverse	Comm ⁿ	Algebraic Structure
1.	$a+b$	✓	✓	✓	✓	✓	Abelian Group
2.	$a \cdot b$	✓	✓	✓	X		Monoid
3.	$\min(a, b)$	✓	✓	X			Semi-Group
4.	$a-b$	✓	X				Groupoid
5.	$a+b-ab$	✓	✓	✓	X		Monoid
6.	a^b	✓	X				Groupoid

$$1. \quad e=0, \quad a^{-1}=-a$$

$$2. \quad e=1 \quad a^{-1}=\frac{1}{a} \quad \forall a \neq 0, \quad a^{-1}=\infty$$

$$3. \quad \min(a, b)$$

$$\min(a, e) = a = \min(e, a)$$

$$\Rightarrow e = \infty \notin R$$

$$5. \quad e=0$$

$$a * a^{-1} = 0$$

$$a + a^{-1} - aa^{-1} = 0$$

$$a + a^{-1}(1-a) = 0$$

$$a^{-1} = \frac{-a}{1-a} \quad a=1$$

$$a^{-1} = -\infty \notin R$$